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RESEARCH ARTICLE

Event-Based Multi-Task Facial Landmark and Blink Detection

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ABSTRACT Facial landmark detection and blink detection are essential tasks in computer vision, with significant applications in behavioral analysis and human-computer interaction. Traditional frame-based cameras struggle with the high temporal resolution needed for accurate blink detection under challenging conditions. This paper presents a novel approach using event-based cameras, which offer superior temporal resolution and dynamic range. By leveraging the asynchronous nature of these cameras, we capture fine-grained motion information, enabling precise real-time detection of rapid eye blinks and facial landmarks. We propose a multi-task deep neural network that optimizes both tasks simultaneously, enhancing overall system performance through shared learning. Our method achieves state-of-the-art results on a public event dataset with a normalized mean error of 4.76% for five facial keypoints. Using a combined dataset of real and synthetic events, we scale the task from 5 facial landmarks to 98, while attaining a normalized mean error of 3.65%. This 98 landmark model simultaneously detects blinks with an F1 score of 0.905, setting a new state-of-the-art in event-based blink detection. With these findings we demonstrate the potential of event-based cameras for robust, real-time facial analysis.

INDEX TERMS Event camera, facial landmarks, blink detection, neuromorphic vision.

I. INTRODUCTION

Facial landmark detection involves identifying and tracking specific key points on the face. Analysis of these landmarks is instrumental in understanding and monitoring human behavior, offering valuable insights into subtle yet significant expressions and movements. Facial landmarks are used in various applications such as head-pose tracking, facial recognition, and detection of facial expressions and micro-expressions [1], [2]. Traditionally, visible-spectrum cameras that capture images in the Red, Green, and Blue (RGB) channels, as well as near-infrared (NIR) imaging, have been

used for this task. While effective for many applications, conventional frame-based cameras face challenges in capturing rapid facial movements due to limited frame rates.

One such rapid motion is blinking, which has a well-established link to drowsiness detection in driver monitoring systems (DMS) [3], serves as an input mechanism for human-computer interaction, particularly for assistive technologies [4], and plays a key role in liveness verification for biometric identity recognition [5]. This makes it a desirable behavior to detect, often accomplished with the help of landmarks around the eyes [6] but is a particularly demanding task in terms of temporal resolution [7]. Given these demands, we explore the capabilities of event cameras. Recognized for their high temporal resolution and ability

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to capture rapid movements with minimal data loss, event cameras offer a promising solution for precise behavioral analysis [8], [9]. Unlike conventional cameras, each pixel in an event camera independently and asynchronously detects changes in light intensity in response to motion. Rather than capturing entire scenes at fixed intervals, event cameras have the potential to detect localized activity at an effective equivalent frame rate exceeding 10,000 frames per second (FPS) and are highly resistant to motion blur [9]. These are critical advantages over conventional frame-based cameras when it comes to the detection of blinks, where the rapid movements of the blink can easily be motion blurred or inadequately captured due to the time between the discrete frames. Additionally, they possess low power requirements and a high dynamic range, enabling their operation in extreme lighting conditions [8]. By only capturing changes in lighting or motion, event cameras reduce the collection of unrelated and redundant data, limiting unnecessary data processing.

Multi-task learning involves training a model on multiple related tasks simultaneously, allowing it to share representations and learn more robust features [10]. In our case, detecting facial landmarks and blinks together provides mutual benefits. Facial landmarks provide crucial context for blink detection, as the positions of the eyes, eyelids, and surrounding areas are directly relevant to identifying blinks. Conversely, blink detection can enhance landmark accuracy, particularly around the eyes, by providing temporal cues that inform the model about the eyes' dynamic state. This shared learning process improves the overall performance and robustness of the system.

This paper presents a novel approach for simultaneous facial landmark and blink detection using event-based cameras. Our method leverages the asynchronous nature of event cameras to capture fine-grained motion information, enabling precise detection of rapid eye blinks and facial landmark positions in real-time. We propose a multi-task deep neural network capable of optimizing both tasks jointly. On a public event dataset with five labeled facial keypoints, our method achieves state-of-the-art performance in event-based facial landmark detection with a normalized mean error of 4.76%. Additionally, using a combined dataset of real and synthetic events, we developed a single model that detects 98 facial landmarks with a normalized mean error of 3.65% and simultaneously detects blinks with an F1 score of 0.905.

In summary, our contributions are threefold:

- 1) We achieve state-of-the-art performance in facial landmark detection on a public event dataset, while increasing the number of landmarks from 5 to 98.
- 2) We introduce a multi-task deep neural network for simultaneous facial landmark and blink detection using event-based cameras, also achieving state-of-the-art performance in event-based blink detection.
- 3) By using a set of simulated events with only landmark labels to improve the performance of both landmark and blink detection on real events, we demonstrate how

synthetic event data can be effectively utilized in a multi-task framework.

These advancements highlight the potential of event-based cameras for robust and real-time facial analysis, paving the way for improved applications in driver monitoring, human-computer interaction, and beyond.

II. RELATED WORK

In this section, we review existing literature on facial landmark detection and blink detection, including both conventional and event-based methods. We follow this with a discussion of multi-task learning approaches in computer vision.

A. FACIAL LANDMARK DETECTION

Facial landmark detection has significantly advanced with the advent of deep learning, particularly through the use of Convolutional Neural Networks (CNNs) [11]. These methods can be broadly categorized into coordinate-regression and heatmap-regression approaches. Coordinate-regression methods directly predict the coordinates of landmarks from an image using CNN-extracted features, typically with a cascaded refinement approach as in [12]. While computationally efficient, these methods often suffer from accuracy issues due to their reliance on visual features alone. In contrast, heatmap-based models such as the one proposed by Zhiwei et al. [13], generate heatmaps representing the likelihood of landmark positions, offering superior accuracy at a higher computational cost.

The Pixel-in-Pixel Network (PIPNet) [14] integrates both heatmap generation and coordinate regression, leveraging their strengths for efficient and precise landmark detection. It employs a novel heatmap-based detection head and a neighbour-regression module to improve robustness. Some works have enhanced facial landmark performance by incorporating other architectures with CNNs, such as Recurrent Neural Networks (RNNs) and attention mechanisms, to better capture the spatial dependencies between landmarks. Xiao et al. demonstrate this with their recurrent model based on a Long Short-Term Memory (LSTM) network modified to include an attention component [15].

B. EVENT-BASED FACIAL LANDMARK DETECTION

Event-based cameras, which capture changes in a scene asynchronously rather than at fixed intervals, offer high temporal resolution and dynamic range, making them suitable for facial landmark detection under challenging conditions. Yu et al. [16] utilized Liquid State Machines (LSMs) to detect visual landmarks around the lips and jaw, aligning them with audio features for improved speech recognition. Savran [17] employed event cameras and CNNs to detect voice activity by tracking facial landmarks corresponding to lip movement. The high temporal resolution enabled accurate detection of subtle changes in lip activity, crucial for robust, real-time voice detection.

Liu et al. [18] introduced NeuroDFD, a driver monitoring system that relies on neuromorphic vision sensors for detecting facial landmarks, adapting to varying lighting and head orientations. Berlincioni et al. [2] used event-based vision to identify key facial landmarks and detect subtle changes in expression, enabling real-time emotion recognition. Kang et al. [19] explored event camera imaging to detect and track pupils for AR/VR systems, using Scale-Invariant Feature Transform (SIFT) features for tracking verification.

All of the above event works are focused on some downstream task that makes use of the landmarks as an intermediate feature. The first and only article we have found to provide accuracy measures for event-based facial landmark detection is by [20], which reports performance on five landmarks and publishes the associated dataset. Showcasing the enhanced capabilities of event cameras in a comprehensive, multi-task learning framework, our work builds on this foundation by expanding the number of detectable landmarks, improving the landmark accuracy, and integrating blink detection,

C. BLINK DETECTION

Blink detection is critical for applications such as driver monitoring and fatigue detection. Traditional methods include frame differencing and optical flow, which, although effective, struggle under low-light conditions and rapid lighting changes [21], [22]. Later works such as [23] and [24] use motion vector analysis over detected eye regions to improve on past methods in both robustness and accuracy. In more recent years, blink detection has increasingly utilized deep learning methods. CNNs have been employed in works such as [6] which uses facial landmarks to obtain the eye-aspect ratio between height and width of the eye and detect blinks. Others have used the features output by CNNs as inputs to LSTMs to take advantage of the temporal cues of blinks [25], [26]. The ongoing surge in popularity for transformer-based networks has also extended to video-based blink detection by [27] and later with a dual-embedding strategy by [28] to achieve state-of-the-art blink detection accuracy, though they suggest that real-time applicability of their pipeline may be limited in its current form.

D. EVENT-BASED BLINK DETECTION

The high-speed requirements and implicit motion in blink detection make it well suited to exploit the unique properties of event cameras. Lenz et al. [29] demonstrated the utility of event-based cameras for face detection and recognition through the dynamics of eye blinks, highlighting the effectiveness of these cameras in capturing rapid eye movements. Ryan et al. [30] extended this application to real-time face and eye tracking, integrating these technologies into systems that greatly benefit from the high temporal resolution and low latency of event cameras. Chen et al. [31] utilized

event cameras to develop NeuroBiometric, an eye blink-based biometric authentication system, showing the potential of event cameras in accurately capturing blink dynamics for secure authentication. These studies indicate the promise of event cameras in enhancing blink detection accuracy and reliability.

While transformer-based architectures have been explored for video-based blink detection [27], [28], their application in event camera systems remains largely unexplored. In a broader event-based vision context, some recent works have begun applying transformer models to event streams. Sabater et al. introduced the Event Transformer [32], which employs a sparse-aware attention mechanism to efficiently process event data, demonstrating strong performance in gesture recognition tasks. More recently, Peng et al. proposed the Group Event Transformer [33], which incorporates group-wise attention for event-based object tracking and classification. Given the effectiveness of CNN-based models in prior event-based facial analytics research [30], [34], we opted to build upon these established architectures for our blink detection system. This choice allows us to leverage proven methodologies while ensuring computational efficiency suitable for real-time applications. Future work could explore the adaptation of transformer-based models to event-based blink detection to assess potential performance improvements.

E. MULTI-TASK LEARNING

Multi-task learning involves training a single model on multiple related tasks, allowing for shared representations and improving overall performance. While not limited to computer vision applications, this approach has been successfully applied in facial analysis [35]. Zhang et al. [10] optimized facial landmarks on RGB images simultaneously with several related tasks, including head-pose, expression (smiling), and wearing glasses. To achieve this, they devised a task-constrained loss function which allows the errors of related tasks to be back-propagated jointly. Working with event data, [34] also utilized head-pose and occlusion (glasses and face masks) tasks in their multi-task training pipeline to be optimized with eye-gaze estimation. In a recent work, Geng et al. [36] introduced an event-based, visible, and infrared fusion system that employs a multi-task collaborative framework. This framework simultaneously performs event-based visible texture reconstruction, event-guided infrared image deblurring, and visible-infrared fusion, showcasing the potential of multi-task learning to enhance image fusion quality in challenging scenarios. Our work extends the application of multi-task learning to simultaneously detect facial landmarks and blinks using event-based cameras. The tasks are inherently related, as the accurate identification of eye landmarks can facilitate better blink detection, and vice versa. This shared learning allows the model to leverage common patterns and representations, leading to more robust and accurate predictions.

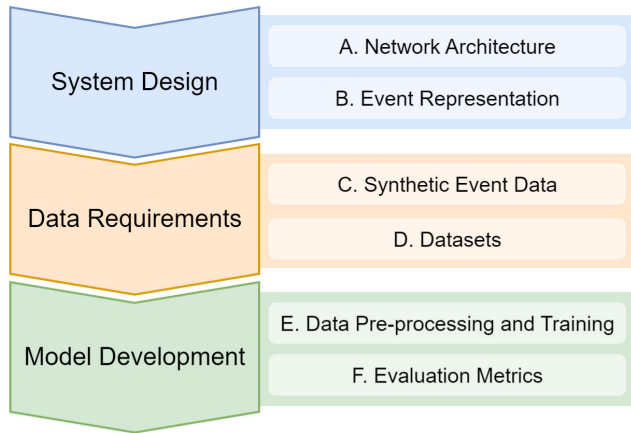


FIGURE 1. Overview of methodology structure.

III. METHODOLOGY

In this section we first provide details on the network design, followed by the approach we took in representing the event data to be compatible with the network. Next, we discuss how RGB and NIR datasets can be converted into synthetic events, before presenting breakdown of all the datasets - both real and synthetic - that were used. We outline some final preprocessing of the data and training details, then finish the section by defining some evaluation metrics used to assess the network's performance in both the landmark and blink detection tasks. An overview of the methodology structure is given in Fig. 1, where the subsection headings have been categorized.

A. NETWORK ARCHITECTURE

This network in this paper is adapted from PIPNet [14] to take advantage of the benefits outlined in Section II-A. The robustness of the model is of particular relevance with event data, as a lack of head motion causes a rapid decline in visibility of the head. The PIPNet authors found MobileNetV2 to be a suitable and lightweight backbone in RGB. With their results and our previous success in using MobileNetV2 to extract facial features from events for yawn detection [37], we selected it as the default backbone in this paper. As with the yawn detection network, the first convolutional layer of the MobileNet was replaced to accept a single-channel input instead of 3-channel.

We modified this PIPNet to include a convolutional LSTM module adapted from [38] to further improve robustness of the landmarks on event streams, and an additional detection head for blinks. By jointly learning on landmarks and eye blinks the model can better understand the interdependencies between the two tasks, which should lead to more accurate and robust performance. A high-level view of the resulting network architecture is shown in Fig. 2. The number of channels N corresponds to the number of facial landmarks output by the network and C denotes the number of neighbouring landmarks predicted. We set C to 10, the

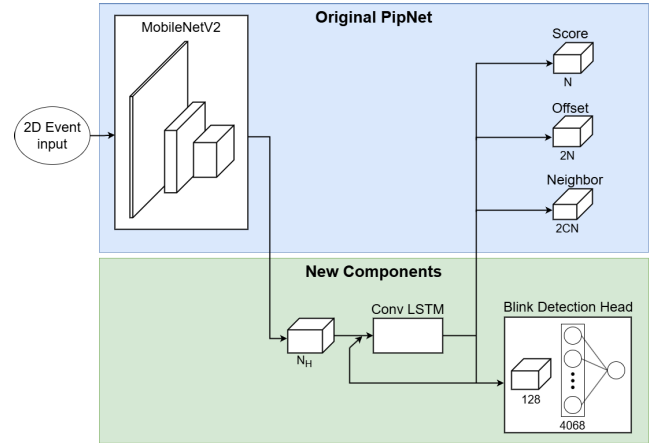


FIGURE 2. Diagram of the adapted PipNet, including a convolutional LSTM module and a blink detection head.

value determined optimal by [14] for an N of 98. The feature maps output by the MobileNetV2 backbone has 1280 channels, which in the original PIPNet are passed directly to the score, offset and neighbour heads. However, to maintain these channel sizes through the convolutional LSTM (comprised of 3 layers) would increase the parameter count by more than a factor of 10. Instead, the feature maps are reduced to hidden channel count N_H before input to the LSTM and subsequent blink detector and original PIPNet heads. A baseline N_H of 512 was selected, setting the parameter count to 10.9M. Experiments with varying values of N_H can be found in Section IV-E of this paper, exploring the complexity requirements for the landmark and blink tasks to determine potential applications in hardware-limited systems.

B. EVENT REPRESENTATION

In an event stream, each event is defined as $e_n = (x_n, y_n, t_n, p_n)$, where x_n, y_n are the spatial coordinates of the event, t_n is the event's timestamp, and p_n is the event's polarity, which has a value of $+1$ or -1 indicating whether there was an increase or decrease in brightness respectively. To be compatible with pre-existing CNN architectures, such as PIPNet, the raw events are aggregated into image-like representations. A common method for representing event data is the histogram approach, where events are grouped into time windows and their polarity values are summed. This can be done using either a fixed number of events per window or a fixed time duration [8]. Formula (1) gives the calculation of P_{hist}^k , the value of a single pixel in an event histogram accumulated over a time window k containing N events. Within the event window k , the subset of events that occur at pixel coordinates (x, y) is denoted by $\mathcal{E}^k(x, y)$.

$$P_{hist}^k(x, y) = \sum_{e_n \in \mathcal{E}^k(x, y)} p_n \quad (1)$$

Defining the windows by a fixed duration preserves more temporal information than a fixed event count. This is useful



FIGURE 3. A comparison of two approaches for representing event data. (a) This histogram representation accumulates events within 33ms windows, showing only recent activity. (b) This time surface representation retains past events with an exponential decay factor of 0.2, improving visibility of features that have stopped moving.

in detecting features with a characteristic temporal signature, such as blinks [30]. It also must be considered that using a fixed duration makes no guarantee of how many events may occur over said duration, and so they are prone to generating images with low spatial information when there is little motion in the scene. This is less of a concern in blink detection specifically, as movement of the eyelids is required for a blink to occur. For this reason, fixed duration histograms are the first representation tested in this article, however, for the detection of facial landmarks we do not have the same assurance of motion. We still wish to detect landmarks on a stationary face, but it can be completely invisible with this representation. This is mitigated by the LSTM addition to the network. With memory of previous inputs, it can learn to distinguish cases where no face is present from cases where the face simply hasn't moved.

As another countermeasure to low facial visibility, we test a second event representation known as a time surface. Like the histograms, our implementation of the time surface uses a fixed duration window of events to generate each input. Where they differ is in the treatment of past events. In the construction of consecutive histograms, the events from previous windows are completely discarded, whereas in a time surface they undergo exponential decay before accumulation. This makes the pixel intensity, calculated according to (2), a function of the age of the events as well as the sum of their polarities. The rate of decay is controlled with τ . With a suitable decay rate we can dramatically increase the visibility of low-motion components by extending the life of each event, as demonstrated in Fig. 3 where the same event stream is represented by both methods. A risk with this approach is past events can introduce some ambiguity on the position of the face if there is small, but sufficient motion in the current event window to generate a suitable histogram. With this reasoning in mind both representations were tested for their relative suitability in the facial landmark detection and blink detection tasks.

$$P_{tsurf}^k(x, y) = P_{hist}^k(x, y) + P_{hist}^{k-1}(x, y) \cdot e^{-\frac{t_N - t_0}{\tau}} \quad (2)$$

C. SYNTHETIC EVENT DATA

A common challenge in event camera research is the scarcity of large, publicly accessible datasets. To address this, researchers have developed event simulators, such as V2E [39]. This open-source tool generates realistic event data by analysing the log-intensity differences across consecutive frames against a predefined contrast threshold, with added temporal noise and leak events. While designed to operate on RGB videos by converting them to luma frames, we have also demonstrated its ability to generate realistic events from NIR data capable of training models that generalize to real events in previous works [37].

With this context, we give the full detail on all datasets used for this research in the next subsection, as a significant portion of the event data were generated using the V2E simulator on NIR and RGB data.

D. DATASETS

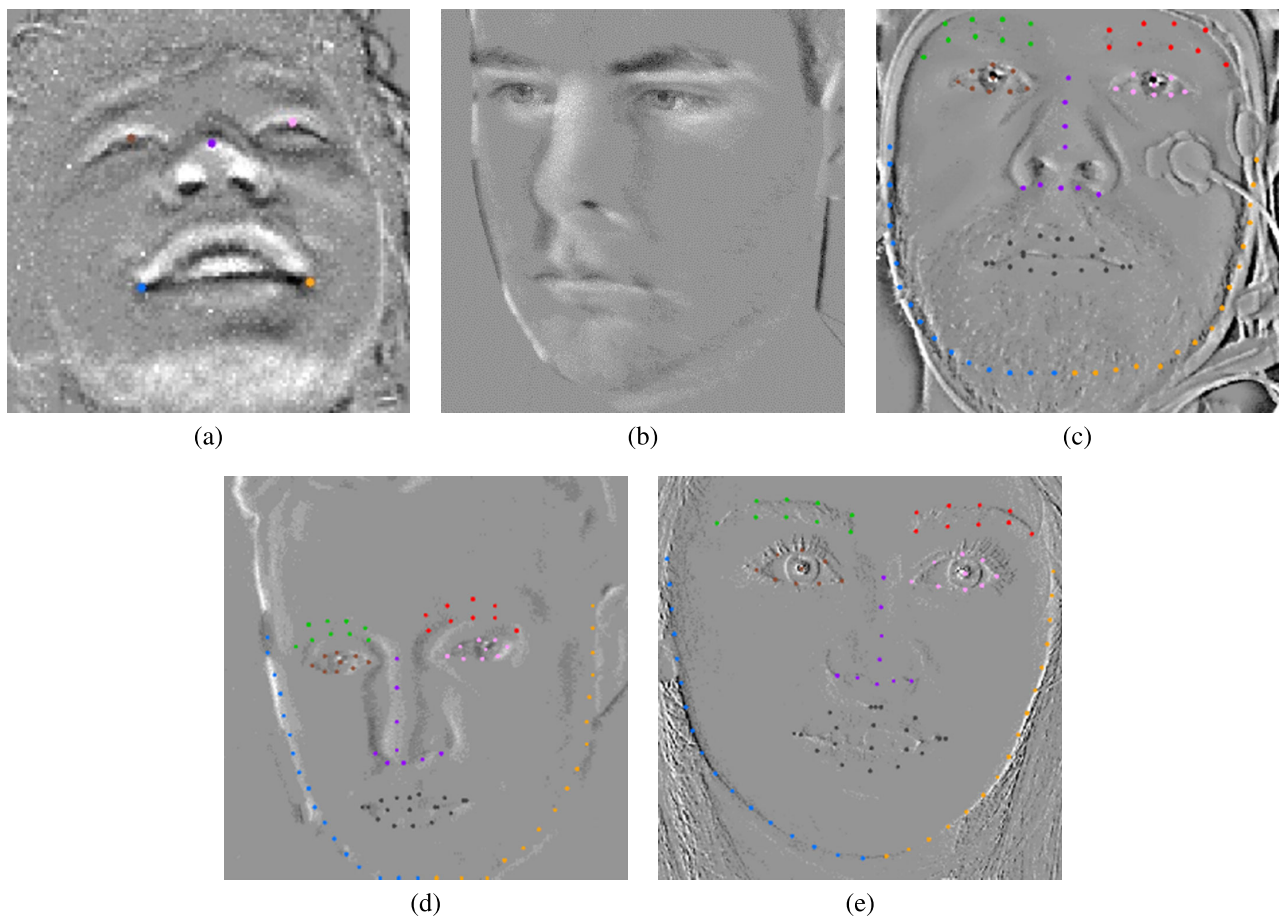
In training and testing the network for both facial landmark estimation and blink detection we drew upon 5 different datasets. In this subsection we discuss the details of each. Real events comprise 2 of the 5 datasets and the remaining 3 use synthetic events. Table 1 compares the features and sizes of the 5 different datasets used in this article and an example time surface from each is displayed with any available landmark annotations in Fig. 4. For the purposes of network training and testing, one sample is defined as an event stream with a duration of at least 0.533s. Longer input durations are theoretically better for more robust predictions by the addition of the LSTM, but also increases training time. 0.533s was chosen as it exceeds the duration of 95% of blink sequences in our data.

1) FACES IN EVENT STREAMS DATASET

The Faces in Event Streams (FES) dataset [20] fills a gap in event research by providing a large and publicly available set of real events with facial landmarks annotations. It is limited to 5 landmarks (eye-centres, nose tip, and mouth corners) but with the authors giving a baseline performance on these landmarks, it is nonetheless valuable for us to test

TABLE 1. Breakdown of features, subject count, and sample count across all datasets.

Dataset	Real Events	Landmarks	Blinks	# Subjects			# Video Samples		
				Train	Valid	Test	Train	Valid	Test
FES (large only + no masks)	✓	✓	✗	40	12	12	2324 (623)	702 (189)	597 (148)
Event-blink	✓	✗	✓	7	3	4	832	180	170
Drowsy-sensors	✗	✓	✓	30	8	8	4191	706	1395
Drowsy-NVS	✗	✓	✗	69	11	16	9639	1613	1806
N-WFLW	✗	✓	✗	-	-	-	1697	300	350

**FIGURE 4.** A sample 33ms event window from each dataset: (a) FES, (b) Event-blink, (c) Drowsy-sensors, (d) Drowsy-NVS, (e) N-WFLW.

the corresponding 5 landmarks in our method. The raw events were accumulated over 33ms windows to create histograms for annotation.

The faces in this dataset are divided into small, medium, and large categories, where large is any face with a bounding box height greater than 90 pixels. This is already smaller

than any of the faces in our other datasets, so our initial experiments used only the large set. We also set aside the 498 video samples that have the participants wearing face masks. The medium, small, and mask wearing samples were reintroduced to the experiments in Section IV-F. Excluded from all experiments and sample counts were a further

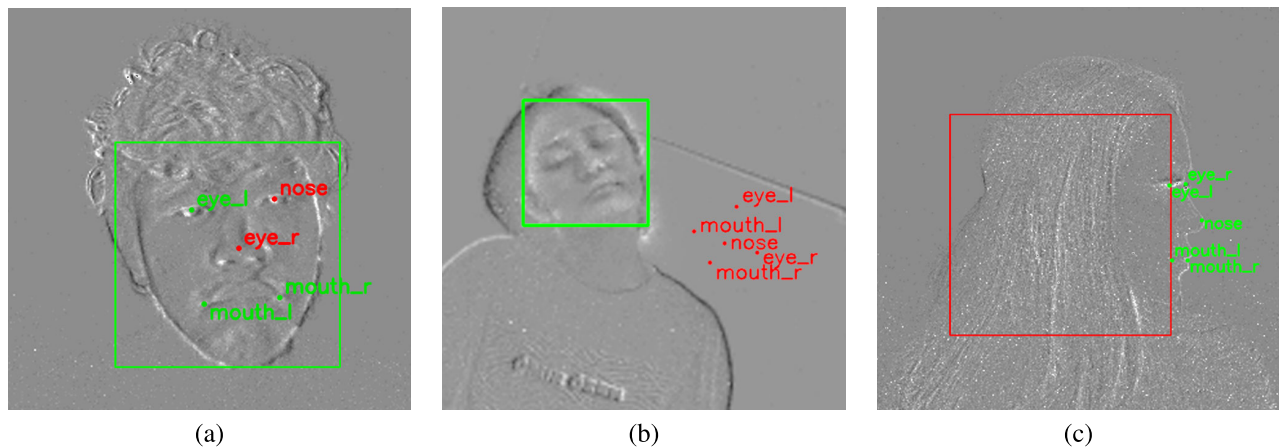


FIGURE 5. Examples of exclusions from the FES dataset with correct elements coloured green and incorrect elements in red: (a) Inconsistent landmark indexing. (b) Correct bounding box but incorrect landmark positions. (c) Incorrect bounding box but correct landmark positions.

116 videos where we observed errors in the labels. The most common of these errors is inconsistent indexing of landmarks, as demonstrated in Fig. 5 (a) where the right eye (as seen by the camera) and nose indices are swapped. Fig. 5 (b) highlights a different error, where the facial bounding box is correct but all landmarks are misplaced outside it. The inverse of this can also occur, as seen in Fig. 5 (c) where the landmarks are correct but the bounding box is not correctly enclosing the face. The full list of videos identified with incorrect labeling will be made available in the supplementary material.

2) EVENT-BLINK DATASET

A smaller local dataset consisting of 14 subjects was collected using a 1280×720 Prophesee event camera. The recordings were focused directly on the face. Blinks were manually annotated over events accumulated every 20ms. Each 20ms event window was given a binary label denoting if it contains eyelid motion that is part of blink. Manually annotating facial landmarks is considerably more time consuming and so this dataset contains only blink labels.

3) NEAR-INFRARED VIDEO DROWSINESS DATASETS

For large-scale data with annotations of 98 facial landmarks, two proprietary NIR datasets focused on capturing drowsiness were included in this work. One dataset features 46 participants wearing a variety of health sensors and includes blink annotations by an electrooculography sensor placed around one eye. The second has 96 unique subjects without any facial sensors but has no blink annotations. In this paper, these sets will be referred to as “Drowsy-sensors” and “Drowsy-NVS” (no visible sensors) respectively. The 30FPS NIR videos were converted to event streams using the V2E simulator, set with contrast threshold of 0.18 to trigger positive and negative events. These recordings were captured with the participants using a driving simulator at set times over a 24 hour period, including 9AM, 5PM, 11PM, and

4AM. This results in diverse lighting conditions across the dataset, though artificial lighting was used for the night-time sessions.

4) WIDER FACIAL LANDMARKS IN-THE-WILD DATASET

The publicly available Wider Facial Landmarks in-the-wild (WFLW) dataset [40] is adopted to create a synthetic neuromorphic-WFLW (N-WFLW) dataset. The WFLW dataset consists of static images with 98 facial landmarks with a wide range of poses and occlusions. However, to simulate synthetic events, we first require videos or image sequences given that events are activated as intensity changes, often through motion. Thus, we generated synthetic video frames by a simulated camera with 6 degrees of freedom over the image on a plane. The resulting frames are cropped equally to the largest dimensions for which the face will always be in bounds. The same transformations are applied to the landmarks for corresponding ground truth labels for each simulated frame. V2E is then used to generate synthetic events from these sequences of images. The feasibility of this method to generate useful data has been previously demonstrated in [30] and [34]. The WFLW annotations contain blur and occlusion flags, both of which were included in our N-WFLW dataset.

E. DATA PRE-PROCESSING AND TRAINING

Each 0.533s event stream is subdivided into windows of equal duration that will be consecutively input to the network. With the landmarks of 3 of our datasets annotated at 30FPS, we chose an event window duration $1/30$ s. This yields 16 consecutive event windows per data sample that are used to form histograms or time surfaces. The time-surfaces use a τ of 0.2, where the units of all event times t in 2 are seconds. Up to 5 event windows directly preceding the desired 16 are processed to “warm up” the time surface. We previously described how event cameras have the potential for much higher equivalent frame rates so it is unfortunate to limit our

network predictions to 30Hz, but in the case of real event data this does not pose the same undersampling risk as a conventional camera running at the same frequency. We can still benefit from the event camera's high sampling rate when accumulating and processing in 1/30s windows. If there is significant blinking motion in the window it will manifest as a wave of events in the histogram/time surface.

All inputs are cropped to a square by the largest dimension of the face bounding box. The training sequences were augmented with a 50% chance of vertical flipping, random rotations of up to 10 degrees, a random cropping and zooming of $\pm 10\%$. Augmentations are only randomized between sequences, so all 16 surfaces undergo the same transformations. For the N-WFLW dataset only, a portion of the event windows were replaced with random noise events to simulate event recordings with no motion. The accumulated event images were clipped to ± 8 before normalization, otherwise a single pixel with a large number of events can suppress all other features in the histogram/time surface.

Training was conducted using mini-batches of 4 sequences, where each sequence comprises 16 consecutive event windows. To balance computational efficiency and learning stability, we accumulated gradients every 16 batches before updating the model weights. The learning rate was initialized at 0.001 and decayed by 20% every 5 epochs over a total of 100 epochs. Weight decay was set at 0.0005 to prevent overfitting.

F. EVALUATION METRICS

The normalized mean error (NME) is used to assess landmark accuracy as in [14], [20]. NME is widely used in facial landmark detection tasks, both in frame-based and event-based methods, as it normalizes the Euclidean error based on facial dimensions, ensuring scale-invariant evaluation. The event-based facial landmark detection benchmark we compare against also reports NME, making it the most appropriate metric for our study. The formula for the NME of a single landmark is given in (3), where the Euclidean distance between the predicted landmark $\hat{x}$ and ground truth x is divided by a normalization distance D . We use the inter-ocular distance for normalization as in [14] for all 98-landmark datasets. The NME on FES is separated from the other datasets as its 5 landmarks don't include those required to calculate the inter-ocular distance for normalization. Instead, we follow the FES authors and normalize the error by $\sqrt{wh}$ where w is the width and h is the height of the face bounding box [20]. Unless otherwise stated, future references to NME represent the average NME of all 98 landmarks on the final event window of each video sample. Our testing requires re-initializing the LSTM layers for every sample, but in real-world usage this would only be required once for single continuous session of any duration. With these sessions unlikely to be as short as our 0.533s testing sequences, we can get a more realistic measure of practical network performance by evaluating on the final event window only. This hypothesis

is tested in Section IV-B.

$$NME = \frac{\|\hat{x} - x\|}{D} \quad (3)$$

Standard metrics for measuring performance in blink detection research include precision (4), recall (5), and F1-score (6) [6], [30], calculated from true positive count, tp , false positive count, fp , and false negative count fn . Precision measures the proportion of detected blinks that are actual blinks, ensuring robustness against false positives, while recall quantifies the proportion of actual blinks correctly detected. Since blink detection involves a trade-off between false positives and false negatives, the F1-score provides a balanced measure as the harmonic mean of the precision and recall, making it a more informative metric than accuracy alone. For each window of events input, the network will predict the likelihood that the events belong to a blink. These probabilities are rounded to the nearest integer (0 or 1) to form the final set of predictions from which the precision, recall, and F1 score can be calculated.

$$Precision = \frac{tp}{tp + fp} \quad (4)$$

$$Recall = \frac{tp}{tp + fn} \quad (5)$$

$$F1 = \frac{2 \times Precision \times Recall}{Precision + Recall} \quad (6)$$

IV. EXPERIMENTS AND RESULTS

This section describes the experiments undertaken to evaluate and iteratively improve the network. At each stage, the model is evaluated up to in 3 ways: the NME on all 98-landmark test datasets, the NME on the FES test data (5 landmarks), and the blink detection F1 score on all test sets with blink labels. With these metrics we understand the impact of the blink and LSTM network components, the effect different datasets have when used in training, and draw comparisons to existing works in event-based landmark and blink detection.

A. EVENT REPRESENTATION EXPERIMENTS

Our first experiment required the training of two PIPNet models with LSTM and blink detection. Both models combine the drowsy-sensors, drowsy-NVS, Event-blink, and N-WFLW datasets, but the events are represented by histograms for the first model and time surfaces in the second. The results are compared in Table 2.

TABLE 2. Comparison of model performance using different event representations.

Event representation	98-landmark NME (%)	FES NME (%)	Blink F1 score
Time surface	3.64	4.46	0.873
Histogram	3.95	6.51	0.90

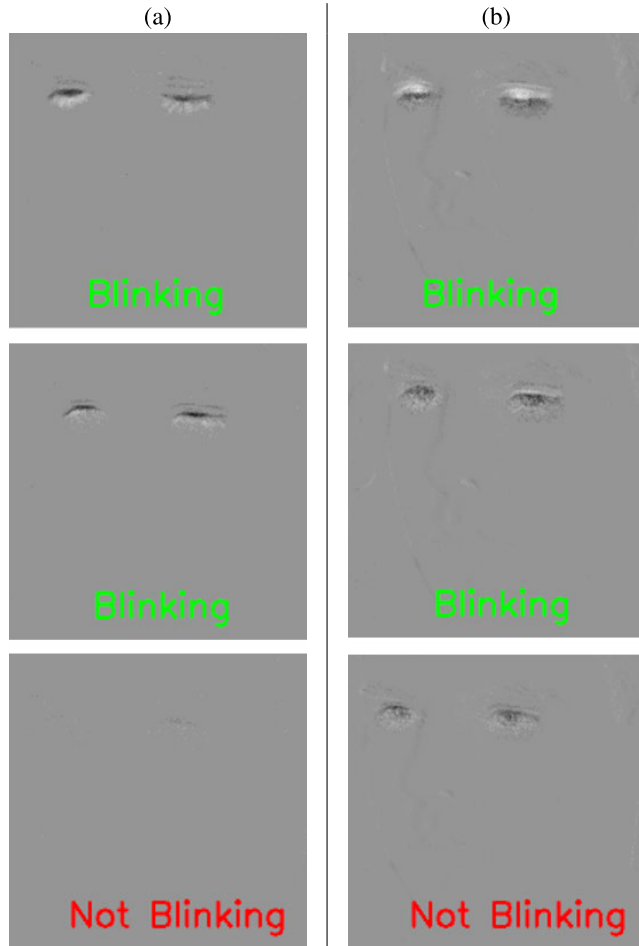


FIGURE 6. $3 \times 33\text{ms}$ windows of an event-blink sequence represented by (a) histograms and (b) time surfaces.

Here we see that histograms hold an advantage in blink detection specifically, though they are a poor choice of representation for landmarks. This is not a surprising outcome, as landmarks require a memory of events in a manner that is not useful for blink detection. A stationary face generates few events but still requires landmark prediction, whereas the eyelid motion that defines a blink will generate the events necessary to detect it.

Furthermore, Fig. 6 demonstrates how the inclusion of past events through a time-surface can obfuscate the exact onset and end point of a blink, where histograms show eye motion in more distinct time windows. However, due to the comparatively high loss in accuracy of the landmarks over the blinks, the time-surface representation was carried forward to subsequent experiments.

B. EXPERIMENTATION ON NETWORK COMPONENTS

To assess the individual contributions of the blink detection head and convolutional LSTM on landmark accuracy, we systematically removed each component before retraining

TABLE 3. Comparison of model performance with blink and LSTM components removed.

Blink head	Conv. LSTM	98-landmark NME (%)	FES NME (%)
✓	✓	3.64	4.46
✗	✓	3.68	4.75
✗	✗	3.77	4.79

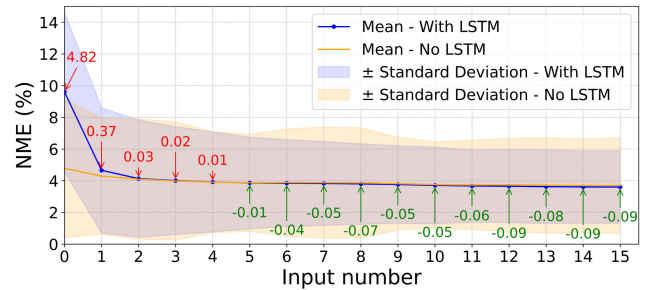


FIGURE 7. An illustration of how the error (NME) of predicted landmarks changes over consecutive inputs. The model with LSTM initially performs worse but improves over time, stabilizing with lower error compared to the non-LSTM model. Annotated on the mean NME for each “With LSTM” input is its difference to the corresponding “No LSTM” mean.

and evaluating a model for each removal. The training data was comprised of the drowsy-sensors, drowsy-NVS, and N-WFLW sets. The Event-blink dataset has just blink labels so it cannot be included when the blink component is removed. The resulting models are compared in Table 3 to the time surface model from the representation experiments in Section IV-A.

The model with both the blink detection head and LSTM achieved the lowest NME across all datasets. The drop in performance that we see when the blink detection head is removed tells us that multi-task learning succeeded in improving the overall accuracy of the landmarks. For all models the error on the FES data is considerably higher than the 98-landmark data, though this gap is smallest when the blink detection head is included, demonstrating more specifically how the joint learning across related facial features improves generalization.

By removing the convolutional LSTM we observe a further increase to the landmark error. To investigate this result more closely we plot the average NME of consecutive input frames for the two non-blink models from Table 3 in Fig. 7. The LSTM model performs significantly worse on first input time surface, caused by the initialization of LSTM memory cells with 0s, but it quickly catches up and surpasses the non-LSTM model from the 6th input. With a consistently lower mean and standard deviation after 0.167s of input data, we believe the LSTM is a valuable inclusion in most practical use cases.

TABLE 4. Comparison of model performance with and without FES data in training and validation sets.

FES in training	98-landmark NME (%)	FES NME (%)	Blink F1 score
$\times$	3.64	4.46	0.873
$\checkmark$	3.65	3.32	0.88

C. EFFECT OF FES DATA IN TRAINING

With just 5 ground truth landmarks, the initial set of models were trained without any FES data, however, for later models the network loss function was modified to expect a landmark mask with each input. This allows the network to predict all 98 landmarks on the FES samples and ignore any that don't correspond to the 5 ground truth landmarks when calculating loss. With all other components kept identical, a new model was trained with FES data included in the training and validation datasets. The change in results on the various test sets are given in Table 4, where we observe a slight NME increase of 0.01 on the 98-landmark test set using the new model. However, this is a very low cost for the vast generalization improvement we see with on the FES NME dropping by 1.14. Despite the FES dataset having no blink labels and only 2 eye landmarks, when included in the training we also see an improvement to the blink detection F1 score.

D. EFFECT OF N-WFLW IN TRAINING

Starting as static RGB images, the N-WFLW dataset is the most abstracted from real events. To investigate the impact these data have on landmark and blink performance and ablation study was carried out. A new model was trained without any N-WFLW data but is otherwise identical to the previous model in which the FES data was added. The results of both models are compared on each dataset in Table 5.

TABLE 5. Comparison of model performance with and without N-WFLW data in training and validation sets.

N-WFLW in training	Landmark NME (%)				Blink F1 score	
	Drowsy sensors	Drowsy NVS	FES	N-WFLW	Drowsy sensors	Real
$\times$	3.65	3.45	3.39	13.1	0.880	0.847
$\checkmark$	3.36	3.26	3.06	6.81	0.882	0.856

We see that the inclusion of N-WFLW grants notably lower landmark error and higher blink F1 scores not just the simulated data but across all datasets, once again conveying the value of synthetic events.

TABLE 6. Comparison of networks with varied parameter count.

N_H	#Params (M)	98-landmark NME (%)	FES NME (%)	Overall NME (%)	Blink F1 score
64	2.6	4.25	3.82	4.23	0.841
256	5	3.76	3.11	3.73	0.85
512	10.9	3.65	3.06	3.63	0.88
768	20	3.73	2.96	3.70	0.857

E. EXPERIMENTS ON NETWORK COMPLEXITY

The number of hidden channels N_H has thus far remained at 512. Additional models were trained to investigate how the channel count affects performance and if another value would be more desirable for either increased accuracy or reduced complexity. The parameter count and performance of these models are compared to the 512-channel model in Table 6.

At approaching double the original number of parameters, the 768-channel model only improves on the 5 landmarks of the FES test set. With a lower error on the 98-landmark and blink test sets, the 512-channel model demonstrates more balanced fitting to the data. To statistically validate whether the observed improvement in landmark performance is significant, we conducted a one-sided paired t-test. The null hypothesis states that there is no difference or an increase in NME when using the 512-channel model compared to the 768-channel model. The alternative hypothesis posits that the 512-channel model produces lower NME than the 768-channel model. Since each test sample has an NME value for both models, a paired t-test was chosen to account for the dependency between paired observations. The test yielded a t-statistic of -3.68 with 3698 degrees of freedom, resulting in a p-value of 0.0012. Since this p-value is well below the standard significance level of 0.05, we reject the null hypothesis and conclude that the 512-channel model achieves a statistically significant reduction in NME compared to the 768-channel model. This suggests that increasing the number of channels beyond 512 does not provide additional benefits and may introduce unnecessary computational overhead, so the focus was shifted to reducing the complexity instead.

With just half of the original parameter count, the 256-channel model is less accurate across the board but presents results competitive enough to warrant pushing the tests further. At 64 channels we see a more significant drop off in performance, setting a floor where the parameter tests were stopped. Aiming for the best overall accuracy we proceeded in this paper with 512 channels, however lighter architectures around the 256-channel mark show potential for use in resource-limited systems.

TABLE 7. Comparison of FES-exclusive networks.

Model	#Params (M)	FES w/ mask NME (%)			
		Small	Medium	Large	Overall
Ours	9.2+12.8*	5.63	4.28	4.50	4.76
DeepFES	24.1	14.69	3.90	1.59	6.35

* Added parameter count of separate face detection network.

F. COMPARISON TO EXISTING LANDMARK DETECTION WORKS

The above experiments do not use any FES videos with face masks or bounding box heights less than 90 pixels. With these exclusions and many other datasets added, it is more difficult to compare our network architecture to the DFES network proposed by the FES authors [20]. Instead, a new model was trained with all of the FES-lab data and the provided train, validation, and test splits. No other datasets were included. This removes the data as a variable and allows a more informative comparison of the architectures. With no blink labels, the blink detection head was also removed. Table 7 compares the parameter count and landmark accuracy by NME of our model and the DFES 33ms model. The faces in this dataset are divided into small, medium, and large categories, where medium is any face with a bounding box height between 35 and 90 pixels. It should be noted that our model is trained on face crops, so the listed parameter count does not include a face detection stage. We have used an event-based face detector from [30] of 12.8M parameters in this paper, though this could be swapped for another network if required, given it outputs a suitably accurate bounding box of the face.

Our method gives a significantly improved overall NME by being more resilient to different face sizes. We also demonstrate improved computational efficiency by reducing the parameter count. The DeepFES model gives a lower error than ours on the large faces, but their change in NME between the different test sizes is 13.1% compared to our 1.35%. We also observe lower error on the medium faces than the large with our method. One contributing factor to this difference is that all of the face mask samples are in the large set. We also observed a higher proportion of extreme head poses in the large set compared to the medium and small sets.

The FES authors do improve upon their results with lower time resolutions of 50ms and 100ms, or respective equivalent frame rates of 20FPS and 10FPS. We did not perform comparable tests as our interests are inclined towards increasing the time resolution to fully draw upon the benefits offered by event cameras.

G. COMPARISON TO EXISTING BLINK DETECTION WORKS

We compare our method with [29] and [30], as they are the only published works on event-based blink detection that

report accuracy metrics. Of these, [30] represents the current state-of-the-art, providing a comprehensive framework for real-time face and eye tracking with blink detection. To better align with how [29] and [30] recorded their results, we modified our approach for evaluating blink detection. Specifically, rather than evaluating blink predictions on a per-event window basis, we counted each unbroken series of blink predictions as a single detected blink. This means a detected blink is considered correct if it overlaps with a ground-truth blink, even if the exact start and stop times differ. This modification ensures that our results are consistent with the evaluation methodology of prior works. With these changes implemented and our results updated, we compare our model to the existing event blink detection algorithms in Table 8.

TABLE 8. Comparison of test sets and results from event-based blink detection works.

Method	# Subjects	# Blinks	Precision	Recall	F1 score
[29]	10	248	-	0.59	-
[30]	3	145	0.869	0.917	0.893
Ours	12	972	0.873	0.937	0.905

From [29], the only metric appropriate for our comparison is the recall, reported as the percentage of blinks that were detected from the total count in a recording. Both precision and recall are presented in [30], from which the F1 score can be calculated. While the metrics are more limited in [29], with 248 blinks across 10 subjects it has a larger and more diverse test dataset than the 145 blinks across 3 subjects of [30]. When considering the fairness of comparing these prior methods to this paper, it should be acknowledged that we have the advantage of much more training data which is likely to contribute to improved generalization and performance. This is balanced somewhat by the increased evaluation complexity that comes with our much larger test set of 972 blinks across 12 subjects.

Our method demonstrates higher precision and recall scores to past event-based blink detection research, with a significantly larger test set. Real events make up 154 of the blink samples, on which an F1 score of 0.916 was achieved. The remaining 818 samples use synthetic events from the drowsy-sensors dataset, and yielded an F1 score of 0.903. We note that across both datasets our model was more inclined to over predict blinks than under predict, with the recall leading the precision by 0.064.

V. CONCLUSION

In this paper, we expand on the limited research of event camera-based facial landmark detection, increasing from 5 to 98 landmarks while reducing the error and network size. This is achieved with a novel recurrent network architecture

adapted from an existing non-recurrent network used to predict facial landmarks RGB images. We additionally demonstrate how blink detection can be incorporated into the network to the benefit of both landmark and blink detection performance through multi-task learning. In developing these models, we made use of simulated event data. Some of these synthetic events started as NIR videos and some as static RGB images, but all were proven beneficial in this article. Our results establish a new state-of-the-art in both event-based facial landmark detection and event-based blink detection.

A. LIMITATIONS AND CHALLENGES

While we have demonstrated that event-based cameras are capable sensors in facial landmark and blink detection, they also present certain limitations that impact their practical deployment. In this section, we discuss the key constraints of event cameras and the challenges associated with their integration into real-world applications.

In applications where significant motion isn't assured, such as facial landmark estimation on a slow moving or stationary face, data sparsity presents a challenge. We mitigate this by leveraging a convolutional LSTM module and a time surface data representation to maintain some spatial consistency over time, but cannot completely eliminate the limitation. This can be further complicated when wishing to process the data for multiple tasks with different requirements. To detect blinks we need to be able to monitor sudden bursts of motion, whereas a facial landmark detector may need to retain much older events to maintain a stable prediction.

Integrating event cameras into existing computer vision contexts presents additional challenges. Event-based data is asynchronous and non-uniform, often requiring unique processing techniques to be used with conventional systems and architectures designed for frame-based inputs. This can introduce additional computational overhead and a potential need for specialized hardware.

Finally, this article and event research in general is impeded by a lack of publicly available benchmark datasets. This is gradually improving as the field matures, but still restricts the rigor of testing and comparing event-based systems.

B. FUTURE WORK

We found that the histogram representation is more suitable for blink detection, while a decaying time-surface representation significantly improves the landmark detection. This provides another avenue for potential future research. We hope to study hybrid event representations or distinct event processing stages as in [30] that could better leverage the beneficial features for both tasks. Through this we could also address some of the limitations and challenges outlined above. A novel processing technique could eliminate the concerns associated with the data sparsity and need for compatibility with conventional systems. This would enhance the applicability of event-based vision systems in practical scenarios.

We also aim to investigate the extent to which our method is scalable to higher time resolutions by training with event windows of shorter durations. We designed the system with this flexibility in mind, meaning it can operate on event windows of any duration and the prediction can be read after any number of windows without modification. This could offer landmark and blink detection at much higher frame rates, making even greater use of the event camera's characteristics. In this paper we were limited to 30Hz by the frequency of our labels (and somewhat by the source data of our synthetic events), but this could be remedied in future works with either new data/labels or interpolation of the existing labels.

Finally, the evidence we have gathered in support of multi-task learning implies the detection of other facial behaviors could add value in a facial landmark network. One such behavior is yawning, for which our research group has previously demonstrated an accurate event-based detection method [37]. With these data and results available, the inclusion of yawn detection is planned to be the basis of our next work.

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